AI Lab - Informed Search Strategies

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Start Your Working Environment

To open the assignment navigate with your browser to: inf-search/inf_search_2_problem.ipynb

Uniform-Cost Search Example

At the beginning of *inf-search/inf_search_2_problem.ipynb* you can find an implementation of the Uniform-Cost Search (UCS), using the graph search strategy. The pseudocode is in the next slide.

Uniform-Cost Search (UCS): graph search version

Require: $problem$
Ensure: $solution$

1: $node \leftarrow$ a node with STATE $= problem.$INITIAL-STATE, PATH-COST $= 0$
2: $frontier \leftarrow$ PRIORITY-QUEUE ordered by PATH-COST, with $node$ as the only element
3: $explored \leftarrow \emptyset$
4: **loop**
5: **if** IS-EMPTY($frontier$) **then return** FAILURE
6: $node \leftarrow$ REMOVE($frontier$) ▷ Remove node with highest priority
7: **if** $problem.$GOAL-TEST($node.$STATE) **then return** SOLUTION($node$)
8: $explored \leftarrow explored \cup node.STATE$
9: **for each** $action$ **in** $problem.$ACTIONS($node.$STATE) **do**
10: $child \leftarrow$ CHILD-NODE($problem, node, action$) ▷ Increase path cost over parent
11: **if** $child.$STATE **not in (** $explored$ **and not in** $frontier$ **)** **then**
12: $frontier \leftarrow$ INSERT($child, frontier$)
13: **else**
14: **if** $child.$STATE **is in** $frontier$ **with higher** $path - cost$ **then**
15: replace that $frontier$ $node$ with $child$

Assignments

- Your assignments for this lesson are at: *inf-search/inf_search_2_problem.ipynb*.
  You will be required to implement some informed search algorithms (Greedy Best First and A*)
- The pseudocodes are variations of the Uniform-Cost Search (UCS) where the *priority queue* is ordered by h and $f = g + h$ respectively

Update the repository

This should not be necessary but if you notice any inconsistency between the code presented by the teacher and the one you have on your local repository, you may need to update the local repository.

Important: do a backup copy of your working directory to make sure you avoid any issue